F79SP Session 9: Markov Chains: Classes

1 Classifying states

Example 1.1. *Reconsider the THMC* $\{X_n\}_{n\geq 0}$ *with* $S = \{1, 2, 3\}$ *with transition matrix* $P = (p_{ij})_{i,j\in S}$.

$$P = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

Observe that:

- *If in state* 1*, must remain in state 1.*
- *If in state* 2*, can go to any state in one move.*
- *If in state* 3*, can go to* 2 *or* 3 *in one move, and to* 1 *in two moves.*

One-way communication

Define the relation $\to$ such that for every $i, j \in S$, $i \to j$ if the following (equivalent) statements holds:

- There exists $n \geq 0$ such that $p_{ij}^{(n)} > 0$.
- There exists a path in transition diagram from state i to j.
- There exists some sequence (possibly empty) of intermediate steps $k_1, k_2, \ldots, k_r \in S$ such that: $p_{ik_1}p_{k_1k_2}\cdots p_{k_rj} > 0$.

Example 1.2. *Reconsider the THMC* $\{X_n\}_{n\geq 0}$ *with* $S = \{1, 2, 3\}$ *with transition matrix* $P = (p_{ij})_{i,j\in S}$.

$$P = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

Observe that:

Intercommunication

Further, define the relation $\leftrightarrow$ such that for every $i, j \in S$:

$$i \leftrightarrow j \text{ if and only if } i \to j \text{ and } j \to i.$$

Note that $\leftrightarrow$ has the following properties.

- Reflexivity: $i \leftrightarrow i$ for every $i \in S$.
- Symmetry: $i \leftrightarrow j$ if and only if $j \leftrightarrow i$.
- Transitivity: $i \leftrightarrow j$ and $j \leftrightarrow k$ implies $i \leftrightarrow k$.

As a result, $\leftrightarrow$ is an **equivalence relation**, which then gives rise to a **partition** of the state space S. The result of this partition are the **classes of intercommunicating states**.

Example 1.3. *Reconsider the THMC $\{X_n\}_{n\geq 0}$ with $S = \{1, 2, 3\}$ with transition matrix $P = (p_{ij})_{i,j\in S}$.*

$$P = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

Observe that:

Open and closed classes

In addition, we can further classify these classes into open and closed using the one-way communication $\to$.

- A class C is said to be **closed** if for every $i \in C$, if $i \to j$, then $j \in C$ (escape from C is impossible).
- Otherwise, the class is said to be **open** (escape from C is possible).

Example 1.4. *Reconsider the THMC $\{X_n\}_{n\geq 0}$ with $S = \{1, 2, 3\}$ with transition matrix $P = (p_{ij})_{i,j\in S}$.*

$$P = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

Observe that:

Some remarks:

- A closed class with single state is called the absorbing state.
- If the entire state space forms a single class, the class is then necessarily closed.

Further Examples

Simple random walk. For the simple random walk with $0 < p < 1$, all states intercommunicate, and so the state space (the integers) is a single, necessarily closed, class.

Suppose now that the state space is restricted to the nonnegative integers and that the state 0 is modified to be *absorbing*. Then the classes and their types are as follows:

Class	Type
$\{1, 2, 3, \dots\}$	nonclosed (open)
$\{0\}$	closed (absorbing)

If instead the state 0 is instead *reflecting*, then once more all states intercommunicate, and so the state space is a single, necessarily closed, class.

Bonus-malus system. The Bonus-malus system introduced earlier had state space $\mathcal{S} = \{0, 1, 2\}$ and transition matrix

$$P = \begin{pmatrix} p & 1-p & 0 \\ p & 0 & 1-p \\ p & 0 & 1-p \end{pmatrix}.$$

It is easier to see that, provided $0 < p < 1$, all states intercommunicate, and so the state space $\mathcal{S}$ is a single, necessarily closed, class.

Health insurance system. The health insurance system introduced earlier had state space $\{H, S, D\}$ and transition matrix

$$P = \begin{pmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ 0 & 0 & 1 \end{pmatrix}.$$

Assume that $0 < p_{ij} < 1$ for $i = 1, 2$, $j = 1, 2, 3$. Then the classes and their types are as follows:

Class	Type
$\{H, S\}$	nonclosed (open)
$\{D\}$	closed (absorbing)

Further example. Take $\mathcal{S} = \{1, 2, 3, 4, 5\}$ and the transition matrix to be given by

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 1/3 & 0 & 1/3 & 0 & 1/3 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then the classes and their types are as follows:

Class	Type
$\{1, 2\}$	closed
$\{3, 4\}$	nonclosed (open)
$\{5\}$	closed (absorbing)